

Question	Marking Guidance	Mark	Comments
10.1	1. Named structures – trachea, bronchi, bronchioles, alveoli; 2. Above structures named in correct order OR Above structures labelled in correct positions on a diagram; 3. Breathing in – diaphragm contracts and <u>external</u> intercostal muscles contract; 4. (Causes) volume increase and pressure decrease in thoracic cavity (to below atmospheric, resulting in air moving in); 5. Breathing out - Diaphragm relaxes and <u>internal</u> intercostal muscles contract; 6. (Causes) volume decrease and pressure increase in thoracic cavity (to above atmospheric, resulting in air moving out);	6	1, 2. Reject mp1 if structures from other physiological systems are named but award mp2 if the correct structures are in the correct order. 4. and 6. For thoracic cavity accept 'lungs' or 'thorax'. 4. and 6. Reference to 'thoracic cavity' only required once. If idea of thoracic cavity is missing or incorrect, allow ECF for mark point 6. 5. Accept diaphragm relaxes and (external) intercostal muscles relax and lung tissue elastic (so recoils).

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10.2	1. Both contain ester bonds (between glycerol and fatty acid); 2. Both contain glycerol; 3. Fatty acids on both may be saturated or unsaturated; 4. Both are insoluble in water; 5. Both contain C, H and O but phospholipids also contain P; 6. Triglyceride has three fatty acids and phospholipid has two fatty acids plus phosphate group; 7. Triglycerides are hydrophobic/non-polar and phospholipids have hydrophilic and hydrophobic region; 8. Phospholipids form monolayer (on surface)/micelle/bilayer (in water) but triglycerides don't;	5 max	All statements must be clearly comparative or linked by the candidate, not inferred from separate statements. Accept mark points shown on adjacent annotated diagrams. 5. Must relate to element. 7. Accept 'non-polar' for hydrophobic and 'polar' for hydrophilic.

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10.3	1. Glucose and galactose; 2. Joined by condensation (reaction); 3. Joined by glycosidic bond; 4. Added to polypeptide in Golgi (apparatus);	4	1. Ignore α or β for glucose
